

HyperDetector: Advanced Persistent Threat Detection via Hypergraph Neural Networks with Enhanced Global Perception

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Abstract

Advanced Persistent Threats (APTs) represent sophisticated cyberattacks that evade detection through stealthy, multistage operations, posing severe risks to critical infrastructure and organizational security. Due to their ability to effectively capture contextual information of attack behaviors, provenance graphs have emerged as a promising approach for APT detection. However, traditional binary edges in provenance graphs fail to represent the collaborative nature of APT attacks, where multiple entities coordinate in single operations, and local graph structures cannot capture the long-range dependencies across attack stages. To address these challenges, we propose HyperDetector, a novel hypergraph-based method for APT detection. First, we introduce hypergraph representation for provenance data, where hyperedges naturally connect multiple entities involved in system events, preserving the higher-order relational structures that characterize APT behaviors. Second, we employ block self-attention mechanisms that enable global reasoning across distant hypergraph regions, effectively linking dispersed attack indicators throughout the system. Through the synergistic integration of these approaches, HyperDetector achieves comprehensive understanding of both localized multi-entity collaborative behaviors and system-wide attack propagation patterns. Extensive evaluations across multiple prominent datasets demonstrate that HyperDetector outperforms state-of-the-art methods, showcasing its effectiveness for robust and holistic APT detection. Additionally, we make our code and datasets publicly available to facilitate reproducibility and foster further research in this critical area.

CCS Concepts

• Security and privacy → Intrusion/anomaly detection and malware mitigation; Network security; • Computing methodologies → Machine learning.

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Keywords

Network Security, Hypergraph Neural Network, Graph Anomaly Detection

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1 Introduction

The sophistication and scale of web-based cyberattacks are escalating at an unprecedented rate, with Advanced Persistent Threats (APTs) causing billions of dollars in losses across cloud infrastructures and web enterprises annually [5, 21, 29]. Modern cyberattacks are no longer confined to simple web defacements but have evolved into highly complex multi-stage campaigns, encompassing command-and-control operations through legitimate web services, cloud supply chain exploitation, and stealthy data exfiltration via encrypted channels. The SolarWinds incident alone compromised over 18,000 organizations through a single web-based update mechanism, while the Colonial Pipeline attack demonstrated how web-facing systems can cripple critical infrastructure [18]. Traditional anomaly detection methods [2, 23], such as signature-based and rule-based intrusion detection systems, have become inadequate in addressing the challenges posed by these sophisticated web-based attacks.

Provenance-based analysis has emerged as a promising approach for APT detection by tracking causality and information flows across distributed systems [4, 15, 33]. By constructing comprehensive activity graphs that capture interactions between web services, APIs, databases, and cloud resources, provenance graphs can reveal attack patterns and trace malicious activities across complex web infrastructures. Recent researchers have proposed various provenance-based detection methods leveraging graph neural networks, statistical analysis, and behavioral modeling. However, these existing methods exhibit critical limitations when applied to modern web environments:

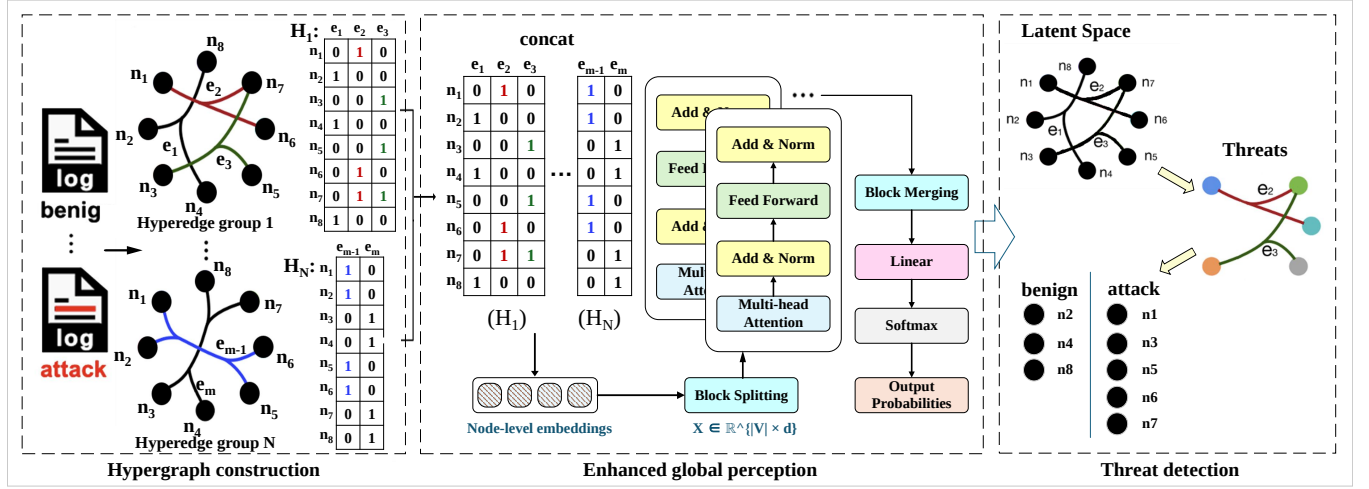


Figure 1: The overall framework of HyperDetector

- **Pairwise Graph Decomposition:** Provenance graphs [23, 26] model system events as pairwise edges between entities, fundamentally losing the multi-entity nature of web transactions. A single API call often involves multiple microservices, databases, caches, and external services simultaneously—this rich context is destroyed when decomposed into binary relationships.
- **Limited Cross-Service Correlation:** Existing methods [19, 32] employ localized message passing or neighborhood aggregation, failing to correlate attack indicators across different web services, cloud regions, and network boundaries. APT actors deliberately fragment their operations across distributed infrastructure to evade such localized detection.
- **Web-Scale Computational Barriers:** Production web environments generate millions of HTTP requests, API calls, and service interactions per hour. Naive global attention mechanisms or exhaustive multi-hop graph traversal [3] become computationally prohibitive, preventing detection at the scale required for modern web applications.

Given these limitations, our core idea is to fundamentally rethink provenance modeling for web environments through hypergraph representation. Unlike traditional graphs, hypergraphs can naturally capture the simultaneous multi-entity interactions inherent in web systems—a single hyperedge can connect all services, resources, and endpoints involved in a complex web transaction. This representation preserves the rich context that APT detection critically depends upon.

We propose HyperDetector, a novel hypergraph-based APT detection framework specifically designed for modern web and cloud environments. HyperDetector addresses the above challenges through two key innovations. First, we construct a web provenance hypergraph where each system event or web transaction becomes a hyperedge connecting all participating entities (web servers, API endpoints, databases, user sessions, cloud resources), preserving the complete context of distributed operations. Second, we develop a hierarchical detection architecture combining Hypergraph Neural

Networks (HGNNs) [8] for capturing local multi-entity semantics with block self-attention mechanisms for efficient global correlation across the entire web infrastructure.

Specifically, we transform web logs, API traces, and cloud audit trails into temporal hypergraph snapshots. For each snapshot, we employ HGNN layers to learn high-order relationships among web components, capturing complex interaction patterns that emerge during APT campaigns such as lateral movement across services or coordinated data staging. To correlate distant attack indicators without prohibitive computational cost, we introduce a block-sparse self-attention mechanism that achieves sub-quadratic complexity while maintaining global receptive fields. The learned representations feed into an unsupervised anomaly detector that identifies deviations from normal web application behavior patterns.

The remainder of this paper is organized as follows: Section 2 surveys provenance-based APT detection, hypergraph learning, and attention mechanisms. Section 3 presents the HyperDetector framework. Section 4 reports experimental results. Section 5 discusses limitations and future work.

2 Related Work

Advanced Persistent Threat (APT) detection has increasingly relied on provenance graphs to model host-level system behaviors. In such graphs, nodes represent system entities, including processes, files, memory objects, and external network endpoints, while directed edges encode causal dependencies (e.g., read, write, exec, connect) extracted from audit logs. Figure 2 illustrates a real-world provenance graph example, where different shapes denote different entity types and directed edges capture their interactions. The upper part of the graph shows benign activities, in which services such as sshd and nginx exhibit regular behaviors, including authentication, configuration loading, and controlled external communication. In contrast, the lower part depicts attacker activities, where a compromised nginx process interacts with untrusted external addresses, performs abnormal executions, accesses sensitive files (e.g., /etc/passwd, /etc/shadow), modifies web resources, and issues

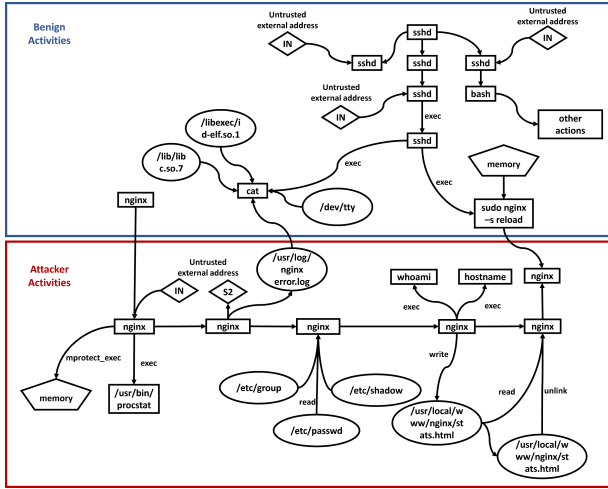


Figure 2: The example of provenance graph.

reconnaissance commands. By explicitly recording fine-grained causal dependencies across processes, files, and network entities, the provenance graph preserves the attack execution context and reveals abnormal dependency patterns that are critical for APT detection and attack reconstruction [20].

Early research on provenance-based APT detection explored various analytical approaches with limited success. Anomaly-based methods like HOLMES [23, 25, 26] and NoDoze [13] analyze anomalous flows and execution paths but suffer from expensive enumeration and noise sensitivity in enterprise environments. Tag propagation methods such as SLEUTH [14] track information flow bidirectionally through decay-based tagging, though they rely on manually-tuned parameters that limit their adaptability. Graph matching approaches including POIROT [22] and UNICORN [12] compare runtime provenance against threat databases but cannot detect zero-day campaigns that deviate from cataloged patterns. These rule-based and pattern-matching systems fundamentally struggle with the dynamic and evolving nature of modern APT attacks.

The advent of deep learning brought neural approaches to provenance analysis, promising to overcome these limitations through learned representations. THREATTRACE [36] pioneered this direction by applying GraphSAGE for learning node embeddings through neighborhood aggregation. Subsequent systems enhanced this foundation: ProGrapher [38] fuses graph embeddings with temporal snapshots, Kairos [4] introduces time-window reconstruction for modeling attack progression, while ShadeWatcher [39] models different edge types through heterogeneous networks. Recent approaches like FLASH [34] and MAGIC [17] incorporate attention mechanisms yet remain constrained to local neighborhoods. Despite these advances, all existing approaches—from early anomaly detectors to state-of-the-art neural methods—share a critical limitation: they operate on pairwise provenance graphs where complex multi-entity events must be decomposed into binary edges.

This decomposition fragments the semantic unity of simultaneous operations—a key characteristic distinguishing malicious from benign behavior.

This pairwise representation fundamentally misaligns with system events that involve multiple entities operating in concert. Consider APT lateral movement: when an attacker’s process simultaneously reads credential files, modifies configurations, establishes network connections, and operates under a compromised user context, current methods must fragment this coordinated action into multiple binary edges. This decomposition forces detectors to reconstruct multi-entity coordination through iterative message passing [10], requiring $O(k)$ aggregation rounds where k represents the coordination distance between entities. The reconstruction process not only introduces computational overhead but also risks losing critical attack patterns where temporal simultaneity signals malicious intent. Furthermore, sophisticated APT campaigns exploit this limitation through time-delayed execution and multi-stage attacks spanning days or weeks [24], creating semantic gaps that neighborhood-based aggregation cannot effectively bridge [1].

Hypergraphs offer a natural solution to this representation mismatch by allowing hyperedge to connect arbitrary numbers of nodes, thereby preserving multi-entity interactions in their original form. In a hypergraph representation, the aforementioned lateral movement scenario becomes a single hyperedge connecting all involved entities—process, files, network socket, and user—maintaining both semantic integrity and temporal correlation. While Hypergraph Neural Networks (HGNN) [8] have demonstrated effectiveness in modeling multi-way relationships across various domains [11, 40], their application to dynamic provenance analysis remains largely unexplored. Existing security-focused hypergraph applications primarily target static analysis scenarios such as malware classification or vulnerability detection, overlooking the critical role of temporal dynamics and simultaneous multi-entity behaviors that characterize APT activities. This gap presents a significant opportunity, as hypergraphs can naturally encode the complex coordination patterns that distinguish sophisticated attacks from legitimate system operations.

Building on this foundation, our analysis identifies three structural limitations in current APT detectors that our hypergraph-based approach directly addresses. First, semantic fragmentation through pairwise decomposition loses the intrinsic atomicity of multi-entity events. In web environments, a single transaction often involves multiple microservices, databases, and external APIs simultaneously—this rich context is destroyed when decomposed into binary relationships, failing to capture critical coordination patterns characteristic of sophisticated attacks. Second, limited cross-service correlation due to reliance on local neighborhood aggregation prevents comprehensive detection across distributed infrastructures. APT campaigns deliberately distribute attack stages across different web services, cloud regions, and temporal boundaries to evade localized detection mechanisms. Third, computational scalability challenges arise when attempting global awareness through traditional message passing in production environments. With millions of HTTP requests and API calls per hour, critical attack signals become diluted through multiple aggregation layers, while exhaustive graph traversal becomes computationally prohibitive for detection. HyperDetector overcomes these limitations by combining

hypergraph neural networks, which preserve multi-entity semantics natively, with block-wise multi-head attention mechanisms that enable efficient long-range reasoning, thereby achieving both semantic completeness and scalable global context understanding essential for detecting sophisticated multi-stage APT campaigns in modern web environments.

3 Method

3.1 HyperDetector APT Attack Detection Framework

HyperDetector detects APT attacks by capturing complex multi-entity relationships and learning system-wide behavioral patterns through two stages: (1) Hypergraph construction, and (2) hierarchical feature learning with global perception, followed by unsupervised anomaly detection.

More specifically, in the first stage, system events from process executions, file operations, and network communications are transformed into a unified hypergraph representation, departing from traditional pairwise graph models. Unlike conventional pairwise modeling, our method constructs higher-order relational structures that capture group behaviors involving multiple entities. Each entity is then embedded into a low-dimensional vector space that preserves both individual attributes and structural positions. In the second stage, hierarchical feature learning is performed through local neighborhood aggregation and global context modeling. Local learning propagates information through multi-entity relationships, while a memory-efficient attention mechanism captures long-range dependencies by dividing entities into blocks and selectively attending to relevant behaviors. This enables detection of both localized and system-wide coordinated attacks. Finally, threat detection employs unsupervised anomaly detection to identify both known and unknown threats without relying on labeled attack data. The system constructs a memory bank of normal behavior patterns from benign system activities. For each new provenance batch, the system analyzes its behavioral patterns and compares them against these established baselines. Through this comparison, entities exhibiting significant deviations from normal patterns are identified as potential threats, enabling the detection of novel attack patterns that have not been seen before.

3.2 Hypergraph Construction

Unlike conventional graph-based approaches that only capture pairwise entity interactions, HyperDetector constructs a hypergraph to model complex multi-entity relationships inherent in APT attacks, as shown in Figure 3. When an attacker compromises a system, they often interact with multiple files, processes, and network connections simultaneously—relationships that simple edges cannot adequately represent.

Hypergraph Construction. We transform heterogeneous security data into a unified hypergraph $\mathcal{H} = (\mathcal{V}, \mathcal{E}, \mathbf{H})$, where $\mathcal{V}$ represents system entities (processes, files, sockets), $\mathcal{E}$ denotes hyperedges capturing multi-way interactions, and $\mathbf{H} \in \{0, 1\}^{|\mathcal{V}| \times |\mathcal{E}|}$ is the incidence matrix. We define:

$$h(v, e) = \begin{cases} 1, & \text{if } v \in e, \\ 0, & \text{if } v \notin e. \end{cases} \quad (1)$$

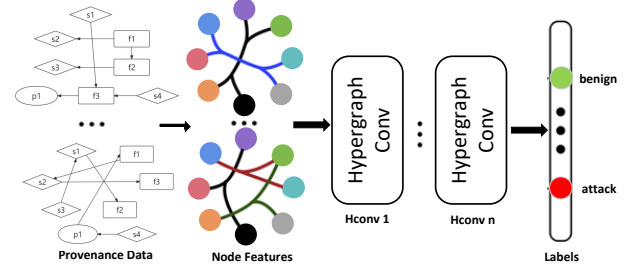


Figure 3: Hypergraph construction from provenance data. Multi-entity system events are preserved as hyperedge, maintaining semantic unity lost in pairwise decomposition.

where each hyperedge $e \in \mathcal{E}$ can connect two or more vertices, with its degree defined as $\delta(e) = \sum_{v \in \mathcal{V}} h(v, e)$.

Given a directed provenance graph $\mathcal{G} = (\mathcal{V}_G, \mathcal{E}_G)$ with temporal annotations, we construct hyperedges by analyzing system audit logs. For each system event that involves multiple entities, we create a hyperedge connecting all participating nodes. Specifically, when a process p performs operations on files $\{f_1, f_2, \dots, f_k\}$ and network sockets $\{s_1, s_2, \dots, s_m\}$ within a single system call or temporal window Δt , we form a hyperedge $e = \{p, f_1, \dots, f_k, s_1, \dots, s_m\}$ with degree $\delta(e) = k + m + 1$.

To enable neural message passing on the hypergraph, we compute the propagation operator $\mathbf{G}$ as the normalized Laplacian:

$$\mathbf{G} = \mathbf{D}_v^{-1/2} \mathbf{H} \mathbf{D}_e^{-1} \mathbf{H}^T \mathbf{D}_v^{-1/2}, \quad (2)$$

where $\mathbf{D}_v = \text{diag}(\sum_e h(v_i, e))$ and $\mathbf{D}_e = \text{diag}(\delta(e_j))$ are the degree matrices of nodes and hyperedges respectively. This formulation preserves the semantic unity of multi-entity operations crucial for detecting coordinated attack behaviors.

3.3 Enhanced Global Perception

Given the hypergraph representation from Section 3.2, HyperDetector extracts node representations through a dual-path architecture that combines:

- (1) **Local high-order modeling** with Hypergraph Neural Networks (HGNN) to aggregate information along hyperedges, capturing attack patterns within multi-entity communities
- (2) **Global context modeling** with a block self-attention module adapted from [27] that efficiently correlates distant system activities

Both components are trained end-to-end, and their complementary nature enables capturing both local malicious patterns and global attack campaigns crucial for APT detection.

(i) *Local High-order Modeling with Hypergraph Neural Networks (HGNN).* Let $\mathbf{X} \in \mathbb{R}^{N \times d_0}$ denote the input feature matrix for N nodes, where d_0 is the dimensionality of initial node features encoding entity attributes (e.g., process privileges, file permissions, network ports). Using the propagation operator $\mathbf{G}$ from Eq. 2, we apply two-layer Hypergraph Neural Networks (HGNN) convolution

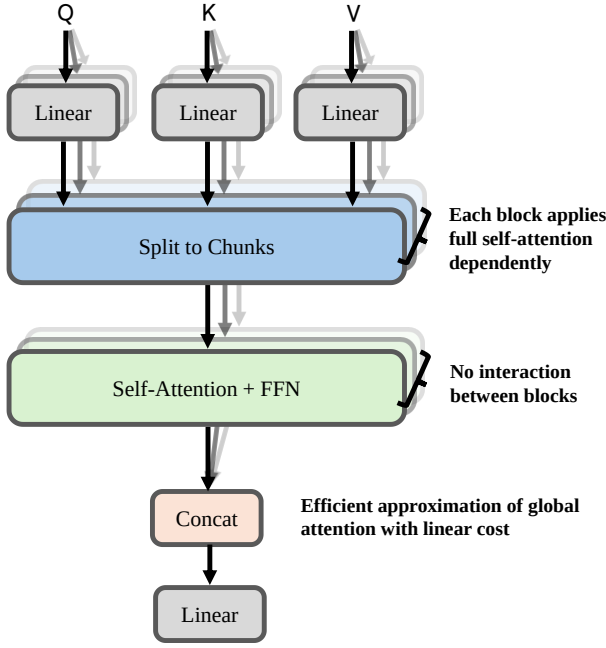


Figure 4: Block-wise Multi-head Attention for efficient global context modeling

to capture multi-hop dependencies:

$$\mathbf{H}^{(1)} = \sigma(\mathbf{GXW}_1 + \mathbf{b}_1), \quad (3)$$

$$\mathbf{H}^{(2)} = \sigma(\mathbf{GH}^{(1)}\mathbf{W}_2 + \mathbf{b}_2), \quad (4)$$

where $\mathbf{W}_1 \in \mathbb{R}^{d_0 \times d_h}$ and $\mathbf{W}_2 \in \mathbb{R}^{d_h \times d_h}$ are learnable weight matrices with hidden dimension d_h , $\mathbf{b}_1, \mathbf{b}_2$ are bias terms, and $\sigma(\cdot)$ denotes ReLU activation. We apply dropout with rate $p = 0.1$ between layers for regularization.

The output $\mathbf{H}^{(2)} \in \mathbb{R}^{N \times d_h}$ encodes high-order interaction patterns propagated through hyperedges. This captures local attack behaviors such as credential dumping tools simultaneously accessing LSASS memory and SAM registry hives, or lateral movement techniques where attackers concurrently establish RDP connections while deploying malicious services across multiple hosts.

(ii) *Global Context Modeling with Block Self-Attention.* While Hypergraph Neural Networks (HGNN) effectively models local multi-entity patterns, sophisticated APT campaigns deliberately fragment their activities across time and system components to evade detection. Standard self-attention could model these global dependencies but requires $O(N^2)$ memory, becoming prohibitive for system graphs with thousands of nodes. To address this scalability challenge, we adapt the block-wise multi-head attention mechanism, as shown in Figure 4.

To address this scalability challenge, we adapt the block-wise multi-head attention mechanisms from [27]. We partition the node sequence into $m = \lceil N/B \rceil$ contiguous blocks of size B (we set $B = 256$ based on empirical evaluation). For block j where $1 \leq j \leq m$, let

$\mathbf{H}_{[j]}^{(2)} \in \mathbb{R}^{B_j \times d_h}$ denote the corresponding node embeddings, where $B_j \leq B$.

Within each block, we apply standard multi-head self-attention [35]:

$$\text{BlockAttn}_j = \text{MHA}(\mathbf{H}_{[j]}^{(2)}, \mathbf{H}_{[j]}^{(2)}, \mathbf{H}_{[j]}^{(2)}), \quad (5)$$

$$\tilde{\mathbf{H}}_{[j]} = \text{LN}(\mathbf{H}_{[j]}^{(2)} + \text{Dropout}(\text{BlockAttn}_j)), \quad (6)$$

$$\hat{\mathbf{H}}_{[j]} = \text{LN}(\tilde{\mathbf{H}}_{[j]} + \text{FFN}(\tilde{\mathbf{H}}_{[j]})), \quad (7)$$

where MHA denotes multi-head attention with $h = 8$ attention heads, each with dimension $d_k = d_h/h$ per head, LN is layer normalization [35], and FFN is a position-wise feed-forward network:

$$\text{FFN}(\mathbf{x}) = \mathbf{W}_4 \cdot \text{ReLU}(\mathbf{W}_3\mathbf{x} + \mathbf{b}_3) + \mathbf{b}_4, \quad (8)$$

with $\mathbf{W}_3 \in \mathbb{R}^{d_h \times 4d_h}$, $\mathbf{W}_4 \in \mathbb{R}^{4d_h \times d_h}$, and bias terms $\mathbf{b}_3 \in \mathbb{R}^{4d_h}$, $\mathbf{b}_4 \in \mathbb{R}^{d_h}$ following the standard Transformer architecture.

This block-wise processing reduces memory complexity from $O(N^2)$ to $O(NB)$, achieving an N/B -fold reduction while preserving full attention within each block. The processed blocks are concatenated to form the final global representation:

$$\hat{\mathbf{H}} = \text{concat}(\hat{\mathbf{H}}_{[1]}, \dots, \hat{\mathbf{H}}_{[m]}) \in \mathbb{R}^{N \times d_h}. \quad (9)$$

This dual-path architecture enables HyperDetector to capture both local coordination (e.g., malware manipulating multiple resources) and global attack flows (e.g., lateral movement across hosts), providing comprehensive coverage essential for modern APT detection.

With these learned representations capturing both local and global behavioral patterns, HyperDetector performs threat detection through unsupervised anomaly detection. We aggregate the learned node embeddings $\hat{\mathbf{H}}$ within each provenance batch to construct compact behavior representations. To model baseline patterns of normal system activities using benign training data, we adopt a generic clustering-based anomaly modeling strategy, where representative benign behavior prototypes are learned; in our implementation, K-means clustering [37] is used as a practical and efficient instantiation, with the number of clusters determined using the elbow method to balance model compactness and representation capacity in a data-driven manner. At inference time, representations from new batches are compared against these benign prototypes, and batches exhibiting significant deviations are flagged as potential threats. This design enables the detection of both known and zero-day APT campaigns without requiring labeled attack data, while remaining flexible to alternative unsupervised anomaly detection techniques.

4 Experiments

4.1 Experimental Datasets

To assess HyperDetector¹, we draw on six open-source provenance graph datasets that together span Android, FreeBSD, Linux desktop, and enterprise-server environments, covering both short-lived and multi-day APT campaigns. All datasets are converted into a unified hypergraph-incidence representation to ensure cross-domain comparability without dataset-specific tuning, and the detailed statistics are shown in Table 1. **APT-2021** [20] captures network flows in a

¹HyperDetector is available at <https://github.com/Fleeterion/HyperDetector>

Table 1: Dataset Description

Dataset	Source platform	V	E	Data size (GiB)
APT-2021	SCVIC corporate network	334 049	518 240	0.72
wget	Debian host (UNICORN-SC)	265 424	975 226	64.00
ClearScope	Android tablet (DARPA TC)	2 309	4 199 309	441.00
CADETS	FreeBSD server (DARPA TC)	59 983	4 811 836	271.00
TRACE	Linux workstation (DARPA TC E3)	3 220 594	4 080 457	15.40
THEIA	Linux server (DARPA TC E3)	1 598 647	2 874 821	17.91

controlled multi-domain testbed simulating realistic APT scenarios across all six kill-chain stages, providing representative class imbalance and diverse attack techniques. **wget** (CVE-2016-4971) records a supply-chain compromise scenario in which a remote-access trojan is embedded within a legitimate software package, comprising 150 three-day provenance graphs that capture complete attack lifecycles. **ClearScope** monitors Android drive-by download and privilege-escalation attacks, offering detailed mobile provenance traces. **CADETS** collects FreeBSD server activities using the Causal Adaptive Distributed Efficient Tracing framework, producing large-scale graphs with millions of edges that reflect enterprise workloads. **TRACE** captures Linux desktop spear-phishing and lateral movement behaviors, representing multi-stage intrusion sequences in realistic host environments. Finally, **THEIA** models enterprise file-share breaches with multi-day activity logs, providing rich semantic annotations of process, file, and network interactions.

4.2 Comparison Methods

To comprehensively evaluate the effectiveness of HyperDetector, we compare it against seven representative state-of-the-art graph-based and hypergraph-based methods that reflect different paradigms in provenance graph analysis for threat detection. These methods encompass various detection granularities (node-level, edge-level, and graph-level) and employ diverse learning strategies, ranging from self-supervised representation learning and anomaly-based detection to inductive neighborhood aggregation and hypergraph modeling.

- **MAGIC** [17]: MAGIC employs masked self-supervised graph representation learning with KD-tree scoring for multi-granularity APT detection.
- **FLASH** [34]: FLASH adopts a GNN-based semantic encoder with Word2Vec-style learning and an embedding cache to support scalable streaming provenance analysis.
- **KAIROS** [4]: KAIROS uses an encoder-decoder GNN to detect edge-level temporal and structural anomalies via graph reconstruction.
- **THREATTRACE** [36]: THREATTRACE performs node-level anomaly detection through inductive neighborhood aggregation for real-time host threat identification.
- **HAG** [16]: HAG models cyber threat intelligence as hypergraphs by extracting multi-entity tactical relations from unstructured CTI reports.
- **HyperGAT** [30]: HyperGAT applies dual-level hypergraph attention to capture high-order behavioral dependencies in provenance graphs.
- **HyperTTC** [6]: HyperTTC leverages hypergraph-based tactic-specific traffic clustering for fine-grained APT detection.

4.3 Comparison Experiments

We conduct extensive experiments to evaluate HyperDetector’s effectiveness against state-of-the-art methods across various attack scenarios. Table 2 presents comprehensive performance metrics across six benchmark datasets, demonstrating HyperDetector’s consistent superiority in diverse attack scenarios and system environments. In addition to conventional GNN-based baselines, we further include recent hypergraph-based methods to evaluate HyperDetector against alternative high-order modeling strategies. Our method achieves the highest F1-scores in 5 out of 6 datasets, with particularly remarkable improvements in challenging long-range detection scenarios where traditional GNN-based approaches struggle.

- **In terms of Precision**, which measures the proportion of correctly identified threats among all predicted ones, HyperDetector achieves the best results on four out of six datasets. It attains 0.914 on APT-2021, 0.938 on wget, 0.997 on ClearScope, and 0.964, 0.971, and 0.985 on CADETS, TRACE, and THEIA, respectively, demonstrating its ability to accurately identify threats with minimal false alarms.
- **In terms of Recall**, which evaluates the proportion of actual threats correctly identified, HyperDetector consistently reaches the highest values across all datasets, with 0.980 on APT-2021, 0.957 on wget, 0.992 on ClearScope, and 0.999 on CADETS, TRACE, and THEIA. This indicates its strong capability in detecting nearly all malicious behaviors while maintaining high sensitivity to rare attack patterns.
- **In terms of F1-Score**, representing the harmonic mean of precision and recall, HyperDetector again delivers the best performance on five out of six datasets. It achieves 0.946 on APT-2021, 0.947 on wget, 0.990 on ClearScope, 0.980 on CADETS, and 0.985 and 0.992 on TRACE and THEIA, respectively, showing balanced accuracy in both detection and reliability. Notably, compared with existing hypergraph-based approaches, including HAG, HyperTTC, and HyperGAT, HyperDetector consistently achieves higher F1-scores across all datasets where results are available, indicating more effective integration of high-order structural modeling and global temporal reasoning.
- **In terms of PR-AUC**, which measures detection quality over all possible decision thresholds, HyperDetector maintains leading or near-leading PR-AUC performance across most datasets, achieving the best results on APT-2021, ClearScope, CADETS, and TRACE. It attains 0.979 on APT-2021, 0.970 on wget, 0.997 on ClearScope, 0.995 on CADETS, and 0.999 and 0.997 on TRACE and THEIA, confirming its robustness and scalability under diverse network environments and threat intensities.

The superior performance of HyperDetector over conventional GNN-based methods arises from its ability to overcome the inherent locality constraints of graph neural networks. Traditional GNNs, while effective at capturing neighborhood-level structures, suffer from local smoothing and message diffusion, which cause information to attenuate across distant nodes and hinder the modeling of long-range dependencies. Even recent hypergraph-based methods,

Table 2: Comparison Experiments. Best results are in bold, second-best are underlined.

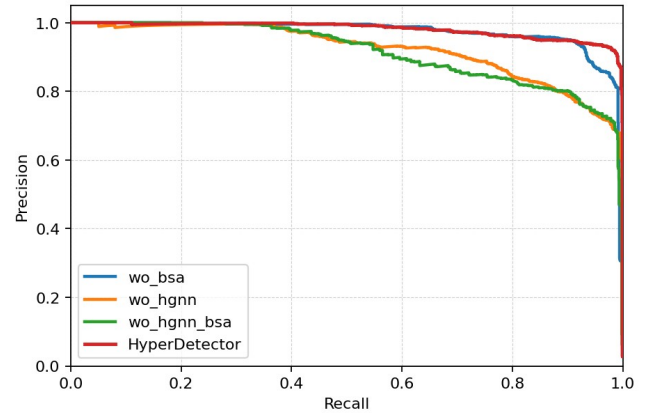
Model	APT-2021				wget				ClearScope				CADETS				TRACE				THEIA			
	Precision	Recall	F1	PR-AUC	Precision	Recall	F1	PR-AUC	Precision	Recall	F1	PR-AUC	Precision	Recall	F1	PR-AUC	Precision	Recall	F1	PR-AUC	Precision	Recall	F1	PR-AUC
MAGIC	<u>0.905</u>	0.810	<u>0.859</u>	0.905	0.941	<u>0.948</u>	<u>0.943</u>	0.971	0.975	0.884	0.917	0.962	<u>0.944</u>	0.988	<u>0.967</u>	<u>0.992</u>	<u>0.970</u>	0.999	<u>0.984</u>	0.999	<u>0.982</u>	0.999	<u>0.990</u>	0.998
FLASH	0.761	0.513	0.612	0.802	0.878	0.902	0.885	0.939	1.000	<u>0.990</u>	0.995	<u>0.995</u>	0.932	<u>0.998</u>	0.959	0.986	0.951	<u>0.988</u>	0.969	0.952	0.918	<u>0.998</u>	0.956	0.966
KAIROS	0.724	0.777	0.750	0.842	0.864	0.852	0.898	0.947	0.891	0.804	0.845	0.845	0.824	0.992	0.942	0.991	0.818	0.999	0.900	0.958	0.803	0.999	0.890	0.947
THREATTRACE	0.885	0.787	0.833	<u>0.910</u>	0.812	0.794	0.823	0.914	0.830	0.774	0.805	0.796	0.716	0.995	0.924	0.957	0.837	0.999	0.911	0.962	0.736	0.999	0.848	0.924
HAG	0.800	0.706	0.750	0.890	0.865	0.891	0.874	0.925	0.834	0.908	0.869	0.942	0.826	0.999	0.904	0.915	0.822	0.999	0.901	0.908	0.624	0.999	0.768	0.956
HyperTTC	0.800	0.588	0.678	0.899	0.902	0.876	0.887	0.929	0.831	0.902	0.865	0.935	0.876	0.999	0.933	0.945	0.820	0.999	0.901	0.921	0.660	0.902	0.763	0.962
HyperGAT	0.784	<u>0.853</u>	0.817	0.881	0.847	0.945	0.892	0.900	0.954	0.876	0.909	0.934	0.931	0.809	0.841	0.952	0.955	0.884	0.918	<u>0.977</u>	0.980	0.939	0.959	0.965
HyperDetector	0.914	0.980	0.946	0.979	<u>0.938</u>	0.957	0.947	<u>0.970</u>	<u>0.997</u>	0.992	<u>0.990</u>	0.997	0.964	0.999	0.980	0.995	0.971	0.999	0.985	0.999	0.985	0.999	0.992	<u>0.997</u>

such as HyperGAT and HyperTTC, primarily emphasize structural aggregation or tactic-level clustering, but still lack explicit mechanisms for modeling long-range temporal correlations across extended attack campaigns. As a result, methods such as MAGIC, FLASH, KAIROS, and THREATTRACE often fail to recognize the subtle, temporally dispersed patterns that characterize APTs. To address these limitations, HyperDetector integrates two complementary components: a Hypergraph Neural Network (HGNN) that models multi-entity interactions through hyperedges, thereby capturing higher-order relations among processes, files, and sockets involved in security-critical events; and a Block Self-Attention (BSA) mechanism that establishes global temporal correlations without being restricted by local message passing. Together, HGNN enriches the structural semantics of provenance graphs, while BSA enables effective propagation of contextual cues across long temporal spans. This synergy allows HyperDetector to accumulate and highlight weak but meaningful behavioral cues over extended periods, leading to more precise and robust detection of complex, multi-stage APT behaviors.

As observed, the APT-2021 dataset presents particularly demanding challenges for detection models. According to the SCVIC-APT-2021 benchmark [20], this dataset realistically represents contemporary network architecture with month-long enterprise traces containing all six stages of APT attacks—from reconnaissance through data exfiltration. The dataset’s complexity stems from APT’s inherent characteristics: well-organized attackers employing diverse techniques, subtle attack patterns that generate limited network flows (leading to severe class imbalance), and temporally dispersed activities designed to evade detection. Despite these heightened requirements, HyperDetector achieves the highest performance with an F1-score of 0.946, dramatically surpassing FLASH by 54.6% and KAIROS by 26.1%. In contrast, hypergraph-based baselines such as HyperTTC and HAG achieve F1-scores of 0.847 and 0.767, respectively, highlighting the advantage of jointly modeling higher-order relations and long-range temporal dependencies. This significant performance gap validates our architectural design—while baseline methods struggle with APT-2021’s month-long traces and sophisticated attack patterns, our hypergraph representation preserves fine-grained attack indicators and the block self-attention mechanism effectively correlates temporally distant events. The dataset authors note that traditional approaches fail when APTs do not follow predicted paths due to their adaptive nature [20], making

Table 3: Ablation results on APT-2021.

Variant	Precision	Recall	F1-Score	PR-AUC
w/o Attention	0.919	0.932	0.926	0.973
w/o Hierarchical	0.810	0.885	0.846	0.924
w/o Hierarchical and Attention	0.778	0.919	0.843	0.917
HyperDetector	0.914	0.980	0.946	0.979

**Figure 5: Precision–Recall (PR) curves of ablation results**

our method’s ability to capture long-range semantic information particularly valuable for detecting these evolving threats.

4.4 Ablation Study

To quantify the contributions of HyperDetector’s key components, we conduct systematic ablation experiments on the challenging APT-2021 dataset, which requires long-range reasoning to detect temporally dispersed attack patterns.

We evaluate four architectural variants:

- **w/o Attention Architecture:** Removes the attention module and relies solely on local hierarchical message passing.
- **w/o Hierarchical Architecture:** Retains temporal attention but replaces the hierarchical encoder with a standard graph-based message-passing network.
- **w/o Hierarchical and Attention Architectures:** A GCN baseline operating on pairwise provenance graphs without hierarchical modeling or attention.

- **HyperDetector**: The complete architecture integrating both hierarchical modeling and attention.

All variants are trained with identical hyperparameters. Table 3 and Figure 5 summarize the results, leading to the following observations.

- **Without the attention architecture**. Removing temporal attention causes a recall drop from 97.9% to 93.2% and an F1 decrease of 2.0%, while precision slightly increases to 0.919. This indicates that the model becomes more conservative without temporal reasoning, missing subtle or temporally distant attack traces. The attention module therefore plays a crucial role in balancing recall and precision by enabling cross-time information aggregation.
- **Without the hierarchical architecture**. Eliminating hierarchical structural modeling leads to clear performance degradation (F1-score 0.846), despite retaining attention. This shows that temporal reasoning alone cannot compensate for the loss of higher-order structural semantics, which are essential for modeling multi-entity APT behaviors.
- **Without both hierarchical and attention architectures**. This variant suffers the most severe degradation, with precision dropping by **13.6%** and F1-score by **10.3%**, although recall remains relatively high. The results indicate that pairwise graphs alone are insufficient to capture the complex dependencies in provenance data, leading to excessive false positives.
- **HyperDetector**. The full model achieves the best performance, with a precision of 0.914, recall of 0.980, F1-score of 0.946, and PR-AUC of 0.979, confirming the effectiveness of jointly modeling hierarchical structure and temporal dependencies.

Overall, the results demonstrate that HGNN and BSA address complementary aspects of APT detection, jointly enabling robust long-range reasoning and balanced precision–recall performance.

4.5 Effectiveness and Scalability of Block Self-Attention

Beyond accuracy, we evaluate the scalability of block self-attention (BSA) under long provenance sequences commonly observed in real-world APT scenarios. We profile inference latency and peak GPU memory usage for sequence lengths N from 1K to 32K on the APT-2021 dataset using an NVIDIA RTX 4050 GPU.

As shown in Table 4, full self-attention exhibits quadratic complexity and becomes impractical as N increases, exceeding 4 GB of GPU memory at 16K tokens and failing with OOM at 32K. In contrast, BSA maintains stable memory growth and predictable runtime across all tested lengths.

Notably, at 32K tokens, BSA requires only 276 MB of GPU memory and 4.07 ms latency, demonstrating near-linear scalability. These results confirm that BSA effectively mitigates the quadratic bottleneck of full attention, enabling HyperDetector to operate on month-long provenance traces without truncation.

By combining block self-attention with hypergraph-based structural modeling, HyperDetector achieves both long-range dependency modeling and practical deployability for large-scale APT detection.

Table 4: Runtime and GPU memory comparison on APT-2021.

N	Full Self-Attention		Block Self-Attention	
	Time (ms)	Memory (MB)	Time (ms)	Memory (MB)
1024	0.307	124.23	0.308	112.34
2048	1.664	173.52	0.337	117.62
4096	5.732	368.08	0.382	128.18
8192	20.307	1141.55	0.922	149.31
16384	79.915	4223.45	1.957	191.56
32768	OOM	OOM	4.074	276.06

5 Conclusion and Future Works

In this work, we introduced HyperDetector, a framework that leverages hypergraph neural networks to preserve multi-entity interactions in provenance data, combined with block self-attention for efficient global reasoning. By capturing high-order relationships directly, HyperDetector overcomes the fragmentation and locality limitations of existing APT detectors. Our evaluation confirms that modeling system events as hypergraphs with global attention is crucial for detecting coordinated threats across temporal and system boundaries.

While HyperDetector demonstrates strong performance and generalization ability, several research avenues remain open. First, we plan to design adaptive mechanisms for handling highly variable or short-term attack patterns that lack explicit multi-entity coordination, enabling finer-grained responsiveness to emerging threat dynamics. Second, domain-specific feature augmentation—such as protocol-aware parsing, semantic labeling of entity roles, and integration of contextual system metadata—will be explored to enhance robustness under sparse or incomplete provenance traces. Finally, we aim to integrate online learning and continual adaptation, enabling evolution alongside emerging APT tactics [7] while maintaining stable performance [9, 28, 31]. In the long term, we envision extending HyperDetector toward cross-system and multi-modal fusion, integrating logs, network telemetry, and behavioral features to build a scalable, self-adaptive defense framework for next-generation enterprise environments.

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